



Simple AWS Reading Lists

Over the last year and a half I've written over a hundred articles, both in my newsletter [Simple AWS](#), and in my personal blog [Cloudy Things](#). I know they're useful, I've referenced them many times as a consultant, both for my clients to dive deeper on a topic, and for myself to remember some details.

But there's so many that it's hard even for myself to remember them all or to find something, even though I've categorized them across domains, and even though I wrote every word. I can only imagine how much worse it must be if you haven't been reading my newsletter since the beginning and want to catch up. So, I decided to put together a few reading lists of my articles, divided by topics.

Keep in mind that these aren't meant to be a complete course on AWS or on any given topic. The articles were written as standalone pieces or part of a short series of 2 or 3 articles. The goal was to dive deeper into some specific topics that came to mind at the time, not to achieve full coverage of any specific domain or topic. They're listed in my recommended order, but each article stands on its own, so you can safely read them in any order, or even skip some.

General topics

These articles are varied in topics, and mostly deal with general stuff that doesn't fit well in the other categories, or that I believe are good to read before diving into the specifics of the other topics. If you're interested in a specific topic such as Containers, scroll down to the other lists.

General topics

[Using AWS Organizations to Manage Multiple AWS Accounts](#) - An introduction to AWS Organizations and why you should use multiple AWS accounts.

[AWS IAM Permissions Explained](#) - A deep dive into how IAM permissions work, and how to grant least privilege permissions.

[Session Manager: SSH to EC2 Instances Securely](#) - A step by step guide on using Session Manager to SSH into EC2 instances without sharing SSH keys.

[Chaos Engineering: Using AWS Fault Injection Simulator](#) - A workshop on how to use AWS Fault Injection Simulator to implement Chaos Engineering practices.

[Why You Should Put API Gateway In Front Of Everything](#) - An opinion piece diving deep into the benefits of using an API Gateway for your APIs. Especially important for Lambda functions, where you typically have multiple distributed services.

[Amazon EBS Basics and Best Practices](#) - An introduction and short discussion on EBS volumes, including tips to optimize cost and performance.

Cloud Architecture

These articles talk about cloud architecture in AWS. It's hard to draw a line between what's cloud architecture and what's cloud infrastructure design, especially when you're talking about the why behind decisions. The criteria that I used for this list is articles that discuss things explicitly from the perspective of cloud architecture. Some of these topics are also covered in articles not on this list, for example Serverless compute is more extensively covered in the AWS Lambda list.

Serverless

[Architecting with AWS Lambda: Architecture Concerns](#) - This article, part 1 of a 3-part series, deals with the thinking behind architecting applications using AWS Lambda.

[Architecting with AWS Lambda: Architecture Design](#) - This article, part 2 of a 3-part series, takes you through the process of architecting a serverless application using AWS Lambda.

[Architecting with AWS Lambda: Best Practices](#) - This article, part 3 of a 3-part series, discusses best practices when architecting applications with AWS Lambda.

Microservices

[Microservices in AWS: Migrating from a Monolith](#) - This introductory article explains the basis of microservices and takes you through a very simple migration from a monolithic application to a microservices architecture.

[Monolith vs Event Driven Architecture with an Example](#) - This article shows you how to build the same application using a monolithic architecture and with an event driven architecture, comparing the two approaches.

Global Infrastructure

[Data Loss, Replication and Disaster Recovery on AWS](#) - This article discusses the importance of replicating data as part of a disaster recovery strategy, analyzing the different data loss scenarios.

[Disaster Recovery and Business Continuity on AWS](#) - This article explains the different disaster recovery strategies available for a global architecture, explaining their key characteristics, advantages and disadvantages, especially costs.

Extras

[High Availability and Scalability in AWS](#) - An opinion piece explaining what high availability and scalability actually mean, how much they cost, and when they're a good idea to apply and when they are a bad one (or rather, an unnecessary one).

[How To Use Amazon SNS to Decouple Components In Cloud Architectures](#) - A brief overview of how to decouple components using events, with Amazon SNS as the event bus.

[Why You Should Put API Gateway In Front Of Everything](#) - An opinion piece diving deep into the benefits of using an API Gateway for your APIs. Especially important for Lambda functions, where you typically have multiple distributed services.

Containers

These articles talk about containers and the different AWS services associated with them. However, they don't cover every aspect of containers. In fact, none of them even actually teach you what containers are! What you will find though is several useful facts and tips about running containers on AWS, especially using ECS and EKS.

Basics

[How to Run Containers in AWS](#) - An introductory article describing a few ways to run containers on AWS.

[Elastic Container Service Basics and Tips](#) - An introduction to Amazon ECS.

[Kubernetes on AWS - Basics and Best Practices](#) - An introduction to Amazon EKS.

[Kubernetes Pods Explained](#) - A brief explanation of what Kubernetes pods are.

[From EC2 to Scalable ECS](#) - A step by step guide to make an application in an EC2 instance scalable and highly available by migrating it to ECS.

[From EC2 to Kubernetes on EKS](#) - A step by step guide to make an application in an EC2 instance scalable and highly available by migrating it to Kubernetes on EKS.

Extras

[CI/CD Pipeline with AWS Code*](#) - A step by step guide to implement a CI/CD pipeline with AWS Code* services for a containerized application running on ECS.

[Why You Should Put API Gateway In Front Of Everything](#) - An opinion piece diving deep into the benefits of using an API Gateway for your APIs. Especially important for Lambda functions, where you typically have multiple distributed services.

[Securing Microservices with AWS Cognito](#) - A guide on how to use Amazon Cognito to secure endpoints (not just microservices) in containerized applications.

AWS Lambda

These articles are focused on AWS Lambda. Some aim to teach you some important concepts about the service, some discuss an architecture approach, and some discuss solutions where AWS Lambda is an important part. I've also included some other articles that don't actually feature AWS Lambda, but talk about concepts or services that I believe are important to know when working with AWS Lambda, such as IAM or API Gateway.

Getting Started

[20 Advanced Tips for AWS Lambda](#) - Despite the title, this article explains a few basics of AWS Lambda, and gives you some high level tips. If you're just starting out you might not be able to fully apply these tips right now, but I recommend you read them anyway, to get a glimpse of what you can and should do.

[Simple Serverless Application with Lambda and DynamoDB](#) - This is an example of the type of application you might build with AWS Lambda, along with a glimpse of the reasoning for choosing a serverless approach.

[Architecting with AWS Lambda: Architecture Concerns](#) - This article, part 1 of a 3-part series, deals with the thinking behind architecting applications using AWS Lambda.

[Architecting with AWS Lambda: Architecture Design](#) - This article, part 2 of a 3-part series, takes you through the process of architecting a serverless application using AWS Lambda.

[Architecting with AWS Lambda: Best Practices](#) - This article, part 3 of a 3-part series, discusses best practices when architecting applications with AWS Lambda.

[Understanding AWS Lambda Pricing](#) - This short article discusses the pricing structure of AWS Lambda.

Advanced

[Managing Multiple AWS Lambda Functions](#) - A discussion on the complexities of managing multiple Lambda functions, and an example of how an application with multiple functions looks like.

[CI/CD for Multiple Lambda Functions](#) - A step by step guide to create a CI/CD pipeline that handles multiple AWS Lambda functions, since you'll most likely have many functions in your application.

[AWS Lambda functions with access to a VPC](#) - If your Lambda functions need to access a VPC, you need to make a few configuration changes. This article will help you understand what you need to do, and give you a step by step guide to do it.

[Observability with AWS X-Ray](#) - A guide on how to implement observability at the code level using AWS X-Ray, including tracing requests across multiple lambda functions.

[Why You Should Put API Gateway In Front Of Everything](#) - An opinion piece diving deep into the benefits of using an API Gateway for your APIs. Especially important for Lambda functions, where you typically have multiple distributed services.

Security

[Monitoring and Securing Serverless Endpoints with API Gateway](#) - A Lambda-specific explanation on how to use AWS API Gateway to monitor and secure the endpoints of your functions.

[AWS IAM Permissions Explained](#) - A deep dive into how IAM permissions work, and how to grant least privilege permissions. Not specific to AWS Lambda, but sorely needed.

[Using AWS WAF to Protect Applications From Common Security Exploits](#) - An article with a deep explanation and step by step guide to using AWS Web Application Firewall to protect your endpoints.

Extras

[Mastering AWS Lambda Cost Optimization: 9 Essential Techniques to Maximize Savings](#) - A few high level tips to reduce your Lambda costs.

[Serverless, event-driven image compressing pipeline with Lambda and S3](#) - An example of an application that you can build using AWS Lambda.

[Best practices for a real-time data processing pipeline with Kinesis and Lambda](#) - Another example of an application built with AWS Lambda.

[Complex, multi-step workflow with AWS Step Functions](#) - Another application that you can build using AWS Lambda.

[Server-Side Rendering with AWS Amplify](#) - Unrelated to Lambda, but a good way to deploy a frontend for your application.

Amazon DynamoDB

These articles all talk about DynamoDB. Full coverage of it was never in my mind when I wrote them, but mostly by accident these articles ended up actually covering most of the important concepts that you need. Don't assume completeness though.

Amazon DynamoDB

[Handling Data at Scale with DynamoDB](#) - This is an introductory article to DynamoDB, explaining what it is, what it does, and what problem it solves. It assumes an intermediate level of knowledge of software in general, and a basic to intermediate understanding of AWS.

[DynamoDB Database Design](#) - An important aspect of DynamoDB is how you design the data that you're going to store. This article covers that in detail, with an example.

[DynamoDB Transactions: An E-Commerce with Amazon DynamoDB](#) - Starting from an example of an E-Commerce application, this article shows you why and how to use transactions in DynamoDB.

[Understanding How DynamoDB Scales](#) - A deep explanation of how scaling works in DynamoDB with Provisioned mode or On Demand mode.

[Reacting to changes with DynamoDB streams](#) - How to use DynamoDB streams to build an event-driven application that reacts to changes in the data.

[Using SQS to Throttle Writes to DynamoDB](#) - An explanation and step by step guide to use an SQS queue to control the rate of write operations to DynamoDB.

Amazon RDS and Aurora

When compiling these lists I figured I'd just make one about databases, and the reason I split them in two was actually DynamoDB and how many articles I've written about it. Imagine how big my surprise was when I realized I actually have written even more about RDS and Aurora! That said, I think this list is even less complete than the others. Many of these articles are from my personal blog, for which I have much lower quality standards. Moreover, some of these articles aren't exactly about RDS or Aurora, but about integrating them with other services like ElastiCache, Client VPN or AWS Lambda. In spite of that, I've personally found many of these articles particularly useful.

Amazon RDS and Aurora

[Managed Relational Databases with AWS RDS and Aurora](#) - An introduction to Amazon RDS and Aurora, and an explanation of what they offer as a managed relational database service.

[AWS RDS Instance Types: Complete Guide](#) - A guide on instance types for RDS. There is some overlap with the next article.

[Understanding AWS Aurora Instance Types](#) - A guide on instance types for RDS and Aurora. There is some overlap with the previous article.

[Aurora Global Database for Disaster Recovery on AWS](#) - A deep dive on using the feature Aurora Global Database to implement a Disaster Recovery strategy for your relational DBs on AWS.

[Ultimate Guide to Understanding and Sizing AWS RDS and Aurora Instances](#) - A short guide on how to resize an RDS or Aurora database instance.

[Using ElastiCache \(Redis\) to cache queries to RDS](#) - A step by step guide on using ElastiCache for Redis to cache queries to RDS.

[Using Client VPN to securely connect to an RDS instance](#) - A step by step guide to set up Amazon Client VPN and use it to connect to an RDS instance without having a Bastion Host.

[AWS Lambda functions with access to a VPC](#) - Step by step instructions on how to configure a Lambda function to connect to an RDS instance.

Security in AWS

I've said this many times before, but it's worth repeating: Security is everyone's job, including yours. And that's even more true if we're talking about security in AWS and you are (or aim to be) the AWS expert. That said, these articles aren't about cybersecurity, and they won't get you anywhere near becoming a security expert. They're about security best practices in AWS that will improve your security posture significantly without requiring you to become a security expert. Ideally, do all of these. And please don't be like me and leave security for the last place.

Must Do

[Using AWS Organizations to Manage Multiple AWS Accounts](#) - An introduction to AWS Organizations and why you should use multiple AWS accounts.

[AWS IAM Permissions Explained](#) - A deep dive into how IAM permissions work, and how to grant least privilege permissions.

[7 Must-Do Security Best Practices for your AWS Account](#) - 7 individual security best practices to improve the security of your AWS account. You don't need to do them all, but some are very easy and very important.

Advanced

[Detecting Failed Sign In Attempts to AWS and Alerting](#) - A step by step guide on how to continuously scan CloudTrail logs to detect and alert on failed sign in attempts.

[Using AWS WAF to Protect Applications From Common Security Exploits](#) - An article with a deep explanation and step by step guide to using AWS Web Application Firewall to protect your endpoints.

[Understanding AWS Pricing: AWS WAF](#) - A short piece on the pricing of AWS WAF.

[Detecting and Fixing Vulnerabilities with Amazon Inspector](#) - An article with an explanation and step by step instructions on using Amazon Inspector to detect and fix vulnerabilities.

Service Specific

[Securing Access to S3](#) - A guide on restricting access to an S3 bucket to only a CloudFront distribution.

[Securing the Connection to S3 from EC2](#) - A guide to set up a VPC Endpoint to access S3 buckets securely from any resource in a VPC (not just EC2 instances, but also ECS tasks, EKS pods, Lambda functions in a VPC, etc).

[Securing Microservices with AWS Cognito](#) - A guide on how to use Amazon Cognito to secure endpoints (not just microservices).

[Session Manager: SSH to EC2 Instances Securely](#) - A step by step guide on using Session Manager to SSH into EC2 instances without sharing SSH keys.

[Using Client VPN to securely connect to an RDS instance](#) - A step by step guide to set up Amazon Client VPN and use it to connect to an RDS instance without having a Bastion Host.